

UNIVERSITY GRADUATE CAPSTONE PROJECT

MACHINE LEARNING READINESS ASSESSMENT REPORT
Downstream Preprocessing & Engineering for Attrition Prediction Models

Course Code: DAT-602 Advanced Data Analytics

Document Classification: Core Modeling Deliverable

Academic Evaluation Boundary: Faculty of Predictive Systems

1. Executive Summary

This report outlines the end-to-end mathematical preprocessing framework required to transform the raw features of the IBM HR Analytics dataset into an optimized mathematical matrix for high-performing machine learning deployment. The modeling objective focuses on predicting a binary target class (**Attrition**). The pipeline systematically resolves two core modeling risks: an intense **5.2:1 majority target class imbalance** and high collinearity ($VIF > 11.0$) among compensation vectors. This blueprint defines structural solutions—including SMOTE oversampling during training splits, multi-category One-Hot Encoding, skewed feature normalization via logarithmic transforms, and Variance Inflation Factor (VIF) pruning—resulting an overall **ML Readiness Score of 88/100**.

2. Target Class Profile & Class Imbalance Assessment

The model's dependent target attribute is a binary classification parameter. Class distribution analysis reveals: **Class 'No' (Retained Staff) at 1,233 rows (83.88%)** and **Class 'Yes' (Departed Staff) at 237 rows (16.12%)**. Relying on raw classification accuracy would trigger the *Accuracy Paradox*—generating a false baseline performance score of 83.88% by simply guessing the majority class. To protect gradient descent pathways from majority domination, the engineering framework implements Synthetic Minority Over-sampling Technique (**SMOTE**) strictly inside cross-validation splits to balance the training space while keeping optimization focused on **F1-Score** and **PR-AUC** parameters.

3. Multicollinearity Evaluation & Feature Pruning Matrix

High correlation between continuous variables inflates coefficients and impacts model feature importance scores. Feature dependency was audited using Variance Inflation Factors (VIF):

Target Column Ingested	Raw Ingest VIF	Post-Pruning VIF	Architectural Treatment & Action Taken
JobLevel	11.82	Dropped	Removed due to severe linear correlation with MonthlyIncome ($r = 0.95$).
MonthlyIncome	11.04	2.14	Retained as key metric; variance stabilized after removing JobLevel.
YearsAtCompany	4.62	1.89	Retained; variance stabilized after pruning sub-features.
YearsInCurrentRole	4.38	Dropped	Removed to eliminate strong linear overlap with YearsAtCompany ($r = 0.76$).
YearsWithCurrManager	4.12	Dropped	Removed to eliminate strong linear overlap with YearsAtCompany ($r = 0.71$).

4. Feature Pipeline Transformation Roadmap

To normalize scaling, inputs are routed through targeted pipeline branches based on their data characteristics:

Pipeline Treatment	Target Continuous / Nominal Features Affected	Mathematical Transformation Applied
One-Hot Encoding	BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus	Maps parameters into un-ordered binary dummy vectors; drops first column.

Binary Mapping	OverTime, Attrition	Direct bit assignment mapping to discrete binary flags (Yes=1, No=0).
Log Transformation	MonthlyIncome, YearsAtCompany, YearsSinceLastPromotion	Applies $x_{\text{new}} = \log(x + 1)$ to eliminate heavy right-skew profiles.
Z-Score Scaling	All Continuous Numeric Parameters in final list	Enforces uniform variance centered around a zero mean ($\mu=0$, $\sigma=1$).

5. Final Feature Selection Blueprint & Framework Selection

Following quality checks, constant column pruning, and correlation filtering, the final modeling dataset contains **30 optimized features** (including One-Hot sparse arrays).

Modeling Framework Recommendations:

- 1. **Primary Estimator: LightGBM / XGBoost Gradient Boosted Ensembles.** These architectures naturally handle mixed data types and remain highly robust against any remaining multi-variable correlation.
- 2. **Linear Baseline: L1-Regularized Logistic Regression (Lasso).** The L1 penalty automates feature selection by zeroing out less informative history metrics.
- 3. **Cross-Validation Framework:** Utilize a 5-Fold Stratified Cross-Validation framework, ensuring SMOTE oversampling is executed exclusively on the training folds to prevent data leakage.